How to solve the integral $\int x^n \sin x dx$?

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Abstract

In this note, we briefly present two approaches for evaluating the integral $\int x^n \sin x dx$: (1) the tabular method for integration by parts and (2) the Feynman's trick which generalizes the original integral to a parameter-dependent integral.

1 Integration by parts

Given two smooth functions f, g of x in an open interval. Let $f^{(0)}, f^{(1)}, f^{(2)}, ..., f^{(n)}$ denote the first n derivatives of f and $g^{(0)}, g^{(-1)}, g^{(-2)}, ..., g^{(-n)}$ denote n antiderivatives of g. For example, $g^{(0)} = g$ and $g^{(-1)} = \int g dx$. The following generalization of the standard integration by parts rule can be obtained using the standard integration by parts successively

$$\begin{aligned}\int fg dx &= fg^{(-1)} - \int f^{(1)} g^{(-1)} dx \\ &= fg^{(-1)} - \left[f^{(1)} g^{(-2)} - \int f^{(2)} g^{(-2)} dx \right] \\ &= ... \\ &= \sum_{m=0}^{n} (-1)^m f^{(m)} g^{(-m-1)} + (-1)^{n+1} \int f^{(n+1)} g^{(-n-1)} dx. \end{aligned} \tag{1.1}$$

The result in Eq.1.1 is particularly useful when one of the functions, let's say f, is given by a polynomial, which means we can always find an integer n such that $f^{(n+1)} = 0$. So the expression above just becomes a finite sum, i.e.

$$\int fg dx = \sum_{m=0}^{n} (-1)^m f^{(m)} g^{(-m-1)}, \qquad \text{if } f^{(n+1)} = 0. \tag{1.2}$$

The result in Eq.1.2 can be represented by a table (so-called tabular method). In the first column we write alternating signs starting from + sign. In the second column we list f and its first n derivatives. In the third column we list g and its first n antiderivatives. We label the columns as D and I to indicate derivatives and antiderivatives. We then multiply diagonally down and to the right to construct the summands of in Eq.1.1, and then we add the products with the corresponding signs attached. By adding the summands, we have

$$fg^{(-1)} - f^{(1)} g^{(-2)} + f^{(2)} g^{(-3)} + \cdots + (-1)^n f^{(n)} g^{(-n-1)},$$

in agreement with Eq.1.2.

Table 1: Tabular method for integration by parts

Signs	D	I	Summands
$+$	f	g	
$-$	$f^{(1)}$	$g^{(-1)}$	$+fg^{(-1)}$
$+$	$f^{(2)}$	$g^{(-2)}$	$-f^{(1)}g^{(-2)}$
$-$	$f^{(3)}$	$g^{(-3)}$	$+f^{(2)}g^{(-3)}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
$(-1)^n$	$f^{(n)}$	$g^{(-n)}$	$\cdots$
	$f^{(n+1)} = 0$	$g^{(-n-1)}$	$(-1)^n f^{(n)} g^{(-n-1)}$

Example 1. $I_2 = \int x^2 \sin x dx$

Signs	D	I	Summands
$+$	x^2	$\sin x$	
$-$	$2x$	$-\cos x$	$-x^2 \cos x$
$+$	2	$-\sin x$	$2x \sin x$
$-$	0	$\cos x$	$2\cos x$

Therefore,

$$I_2 = x^2(-\cos x) + 2x \sin x + 2\cos x$$

Example 2. $I_4 = \int x^4 \sin x dx$

Signs	D	I	Summands
$+$	x^4	$\sin x$	
$-$	$4x^3$	$-\cos x$	$-x^4 \cos x$
$+$	$12x^2$	$-\sin x$	$4x^3 \sin x$
$-$	$24x$	$\cos x$	$12x^2 \cos x$
$+$	24	$\sin x$	$-24x \sin x$
		$-\cos x$	$-24\cos x$

Therefore,

$$\begin{aligned} I_4 &= -x^4 \cos x + 4x^3 \sin x + 12x^2 \cos x - 24x \sin x - 24 \cos x \\ &= (-x^4 + 12x^2 - 24)\cos x + (4x^3 - 24x)\sin x \end{aligned}$$

2 Feynman's trick

Normally, the integral that we want to solve is of the form $I = \int f(x)dx$. The idea of this method is to artificially introduce a parameter so that the integral is parameter-dependent, i.e. $I(t) = \int f(x,t)dt$. This generalized function $f(x,t)$ reduces to $f(x)$ as t approaches a certain value t_0, i.e. $f(x,t_0) = f(x)$. Then the original integral can be obtained by calculating the value $I(t_0)$ as $I(t_0) = \int f(x,t_0)dx = \int f(x)dx = I$. This method is often applied in the following cases:

- $I(t)$ can be related to other integrals that are easy to perform.
- The derivatives of $I(t)$ with respect to t can be easily calculated, and one can build a differential equation in terms of these derivatives. Solving this differential equation allows us to compute the original integral.

To apply the Feynman's trick, we first rewrite the integral in the following way

$$\boxed{I_n = \int x^n \sin x dx = \frac{1}{2i}\int x^n e^{ix} dx - \frac{1}{2i}\int x^n e^{-ix} dx.}$$

Notice that the second term is the complex conjugate of the first term, so we only need to calculate the first term $\frac{1}{2i}\int x^n e^{ix} dx$. For later convenience we denote the first term by I,

$$I = \frac{1}{2i}\int x^n e^{ix} dx.$$

A useful observation is that if we generalize I to be parameter-dependent into the following form

$$I(k) = \frac{1}{2i}\int x^n e^{ikx} dx,$$

then $I(k)$ can be formally calculated from a simple integral $\int e^{ikx}dx$ by differentiating it with respect to the parameter k. Each time we differentiate $\int e^{ikx}dx$ with respect to k, a factor of ix is introduced to the integrand. Consequently, $I(k)$ is related to the n-th derivative of $\int e^{ikx}dx$ with respect to k as follows:

$$I(k) = \frac{1}{2i^{n+1}}\partial_x^{(n)} \int e^{ikx} dx = \frac{1}{2i^{n+1}}\partial_k^{(n)} \left(\frac{e^{ikx}}{ik}\right),$$

where $\partial_k^{(n)}$ denotes the n-th derivative with respect to k. Then we can use the generalized Leibniz rule, namely

$$\partial^{(n)}(fg) = \sum_{m=0}^{n} \begin{pmatrix} n \\ m \end{pmatrix} \partial^{(n-m)} f \partial^{(m)} g,$$

where $\begin{pmatrix} n \\ m \end{pmatrix} = \frac{n!}{m!(n-m)!}$ is the binomial coefficient. Let $f = e^{ikx}$ and $g = \frac{1}{k}$, and notice that $\partial^{(n-m)} f = (ix)^{n-m} e^{ikx}$ and $\partial^{(m)} g = (-1)^m m! \frac{1}{k^{m+1}}$, then

$$I(k) = \frac{1}{2i^{n+2}} \partial_k^{(n)} \left(\frac{e^{ikx}}{k} \right) = \frac{1}{2i^{n+2}} \sum_{m=0}^{n} \begin{pmatrix} n \\ m \end{pmatrix} (ix)^{n-m} e^{ikx} \cdot (-1)^m m! \frac{1}{k^{m+1}}.$$

Inserting $k = 1$ to $I(k)$gives rise to the integral I, then

$$I = \frac{1}{2i^{n+2}} \sum_{m=0}^{n} (-1)^m m! \begin{pmatrix} n \\ m \end{pmatrix} (ix)^{n-m} e^{ix}.$$

Therefore, the original integral I_n thus becomes

$$\begin{aligned} I_n = I + I^* &= \frac{1}{2i^{n+2}} \sum_{m=0}^{n} (-1)^m m! \begin{pmatrix} n \\ m \end{pmatrix} (ix)^{n-m} e^{ix} + \frac{1}{2(-i)^{n+2}} \sum_{m=0}^{n} (-1)^m m! \begin{pmatrix} n \\ m \end{pmatrix} (-ix)^{n-m} e^{-ix} \\ &= \boxed{\frac{1}{2} \sum_{m=0}^{n} (-1)^m m! \begin{pmatrix} n \\ m \end{pmatrix} x^{n-m} \left(\frac{1}{i^{m+2}} e^{ix} + \frac{1}{(-i)^{m+2}} e^{-ix} \right)}. \end{aligned}$$

The expression above can be further simplified using $e^{ix} = \cos x + i \sin x$ and $i^m = \exp\left(im\frac{\pi}{2}\right)$,

$$\begin{aligned} I_n &= \frac{1}{2} \sum_{m=0}^{n} (-1)^m m! \begin{pmatrix} n \\ m \end{pmatrix} x^{n-m} \left[\left(\frac{1}{i^{m+2}} + \frac{1}{(-i)^{m+2}} \right) \cos x + \left(\frac{1}{i^{m+1}} + \frac{1}{(-i)^{m+1}} \right) \sin x \right] \\ &= \sum_{m=0}^{n} (-1)^m m! \begin{pmatrix} n \\ m \end{pmatrix} x^{n-m} \left[\cos\left((m+2)\frac{\pi}{2} \right) \cos x + \cos\left((m+1)\frac{\pi}{2} \right) \sin x \right] \\ &= \boxed{\sum_{m=0}^{n} (-1)^m m! \begin{pmatrix} n \\ m \end{pmatrix} x^{n-m} \left[-\cos\frac{m\pi}{2} \cos x - \sin\frac{m\pi}{2} \sin x \right]}, \end{aligned}$$

where we have used the identities $\cos\left((m+2)\frac{\pi}{2}\right) = -\cos\frac{m\pi}{2}$ and $\cos\left((m+1)\frac{\pi}{2}\right) = -\sin\frac{m\pi}{2}$ in the last line.

Example 3. $I_1 = \int x \sin x dx$: In this case $n = 1$, and m can only be 0 or 1. Then

$$\begin{aligned} I_1 &= (-1)^0 0! \begin{pmatrix} 1 \\ 0 \end{pmatrix} x \left[-\cos\frac{0 \cdot \pi}{2} \cos x - \sin\frac{0 \cdot \pi}{2} \sin x \right] + (-1)^1 1! \begin{pmatrix} 1 \\ 1 \end{pmatrix} x^0 \left[-\cos\frac{\pi}{2} \cos x - \sin\frac{\pi}{2} \sin x \right] \\ &= -x \cos x + \sin x \end{aligned}$$

Example 4. $I_2 = \int x^2 \sin x dx$: In this case $n = 2$, and m can only be 0 , 1, 2. Then

$$I_2 = x^2(-\cos x) + 2x \sin x + 2 \cos x$$

Example 5. $I_3 = \int x^3 \sin x dx$: In this case $n = 3$, and m can only be 0 , 1, 2, 3. Then

$$\begin{aligned} I_3 &= x^3(-\cos x) + 3x^2 \sin x + 6x \cos x - 6 \sin x \\ &= (6x - x^3) \cos x + (3x^2 - 6) \sin x \end{aligned}$$

Example 6. $I_3 = \int x^3 \sin x dx$: In this case $n = 3$, and m can only be 0 , 1, 2, 3, 4. Then

$$\begin{aligned} I_4 &= x^4(-\cos x) - 4x^3(-\sin x) + 12x^2 \cos x - 24x \sin x + 24(-\cos x) \\ &= (-x^4 + 12x^2 - 24) \cos x + (4x^3 - 24x) \sin x \end{aligned}$$